

Document filename: 260126ab
Spectrum ID:
Model ID: Default Model

EXPERIMENTAL CONDITIONS			
Excitation mode: X-ray fluorescence (XRF)			
Sample detector path: Air		Detector type: Si(Li)	
Path length: 3.0 cm		Dead layer: 0.0300 µm	
Primary energy: 16.5 Kev		Contact layer: 0.0200 µm	
Incidence angle: 45.0 degrees		Active depth: 3.0 mm	
Take-off angle: 45.0 degrees		Window: 760.00 µm	
Pulse pile-up resolution time: 2.00 µs			

Fitting Region: channels 103 - 1703, 4 iterations done, last change = 0.03%

Lambda = 1e-006 , ChiSquare = 7.9

CALIBRATION DATA			
	Initial estimate	Final estimate	
ZERO (eV)	-388.7 ± 100.0	-386.4 ± 0.4	
GAIN (eV/ch)	14.664 ± 2.000	14.644 ± 0.002	
NOISE (eV)	62.7 ± 40.0	57.0 ± 3.0	
FANO factor	0.193 ± 0.050	0.134 ± 0.004	

PEAK DATA

# Line	E(KeV) chan#	Rel. int. fwhm (eV)	Peak area backgr	st. dev	Chi-sq tot. abs
1 P K			3888 ± 95		
KA1	2.013	1.00000	3888 ± 95	1.1	
	163.845	94.93	2529		1.49e-005
2 S K			19216 ± 229		
KA1	2.307	0.80932	15552 ± 151	12.8	
	183.921	99.27	4400		5.81e-004
KB1	2.464	0.19068	3664 ± 79	74.9	
	194.642	101.51	5147		2.22e-003
3 Cl K			107995 ± 406		
KA1	2.622	0.82495	89091 ± 303	110.2	
	205.431	103.71	6540		6.32e-003
KB1	2.816	0.17505	18904 ± 103	358.1	
	218.678	106.35	6754		1.66e-002
4 Ar K			17528 ± 381		
KA1	2.957	0.81836	14344 ± 240	156.5	
	228.306	108.23	6677		2.91e-002
KB1	3.191	0.16920	2966 ± 91	29.3	
	244.285	111.29	6179		5.90e-002
KA1 Escape	1.215	0.01046	183 ± 26	8.9	
	109.352	82.02	643		1.92e-022
KB1 Escape	1.449	0.00199	35 ± 24	2.5	
	125.331	86.01	579		1.45e-013
5 K K			18183 ± 329		
KA1	3.313	0.82516	15004 ± 187	25.8	
	252.616	112.84	5790		7.89e-002
KB1	3.590	0.16393	2981 ± 75	52.7	
	271.531	116.30	4490		1.36e-001
KA1 Escape	1.571	0.00925	168 ± 29	3.0	
	133.662	88.02	812		8.15e-011
KB1 Escape	1.848	0.00165	30 ± 39	3.5	
	152.577	92.41	1516		5.92e-007
6 Ca K			12510 ± 314		
KA1	3.691	0.83279	10418 ± 143	11.2	
	278.428	117.54	3922		1.59e-001
KB1	4.013	0.15777	1974 ± 60	3.1	
	300.416	121.40	3024		2.40e-001
KA1 Escape	1.949	0.00809	101 ± 46	3.3	
	159.474	93.96	2128		4.84e-006
KB1 Escape	2.271	0.00135	17 ± 65	13.3	
	181.462	98.75	4240		4.18e-004
7 Ti K			584 ± 301		
KA1	4.509	0.85258	498 ± 79	0.9	
	334.286	127.11	3161		3.66e-001
KB1	4.932	0.14056	82 ± 61	1.6	
	363.171	131.79	3680		4.65e-001
KA1 Escape	2.767	0.00602	4 ± 82	321.6	
	215.332	105.69	6745		1.33e-002
KB1 Escape	3.190	0.00084	0 ± 79	29.3	
	244.217	111.27	6179		5.89e-002

8 Cr K			580	± 287	
KA1	5.412	0.86647	503	± 93	1.9
	395.948	136.90	4305		5.60e-001
KB1	5.947	0.12866	75	± 76	0.9
	432.481	142.38	5669		6.47e-001
KA1 Escape	3.670	0.00434	3	± 64	14.5
	276.994	117.28	4089		1.54e-001
KB1 Escape	4.205	0.00053	0	± 54	1.8
	313.527	123.64	2881		2.89e-001
9 Mn K			944	± 299	
KA1	5.895	0.86610	818	± 105	1.1
	428.930	141.86	5583		6.39e-001
KB1	6.491	0.12983	123	± 81	2.3
	469.629	147.75	6361		7.15e-001
KA1 Escape	4.153	0.00363	3	± 54	1.9
	309.976	123.04	2901		2.76e-001
KB1 Escape	4.749	0.00043	0	± 59	0.8
	350.675	129.79	3451		4.24e-001
10Fe K			18341	± 378	
KA1	6.399	0.86901	15939	± 170	3.5
	463.347	146.86	6262		7.05e-001
KB1	7.059	0.12764	2341	± 86	3.3
	508.415	153.16	6875		7.71e-001
KA1 Escape	4.657	0.00300	55	± 58	0.8
	344.393	128.77	3342		4.02e-001
KB1 Escape	5.317	0.00035	6	± 65	1.7
	389.461	135.90	4182		5.43e-001
11 Co K			1403	± 349	
KA1	6.925	0.87101	1222	± 120	1.1
	499.265	151.90	6777		7.59e-001
KB1	7.649	0.12623	177	± 90	1.7
	548.704	158.57	7971		8.15e-001
KA1 Escape	5.183	0.00248	3	± 63	1.4
	380.311	134.49	3992		5.17e-001
KB1 Escape	5.907	0.00028	0	± 75	1.1
	429.750	141.98	5605		6.41e-001
12Ni K			1266	± 516	
KA1	7.478	0.57714	731	± 106	2.2
	537.027	157.02	7902		8.04e-001
KA2	7.461	0.29531	374	± 93	2.2
	535.866	156.87	7888		8.02e-001
KB1	8.265	0.12527	159	± 91	1.5
	590.768	164.04	8104		8.51e-001
KA1 Escape	5.736	0.00135	2	± 73	1.3
	418.073	140.25	5361		6.15e-001
KA2 Escape	5.719	0.00070	1	± 73	1.5
	416.912	140.07	5316		6.12e-001
KB1 Escape	6.523	0.00023	0	± 80	2.4
	471.814	148.06	6393		7.19e-001
13Cu K			918	± 532	
KA1	8.048	0.57872	531	± 107	1.6
	575.950	162.13	8077		8.39e-001
KA2	8.028	0.29671	272	± 95	1.6
	574.584	161.96	8075		8.38e-001
KB1	8.905	0.12269	113	± 91	1.5
	634.471	169.53	8065		8.79e-001
KA1 Escape	6.306	0.00112	1	± 78	7.2
	456.996	145.95	6139		6.94e-001
KA2 Escape	6.286	0.00058	1	± 78	7.3
	455.630	145.75	6121		6.92e-001
KB1 Escape	7.163	0.00018	0	± 83	4.2
	515.517	154.12	6943		7.80e-001
14Zn K			4489	± 559	
KA1	8.639	0.57753	2593	± 114	0.8
	616.307	167.27	8101		8.68e-001
KA2	8.616	0.29663	1332	± 97	0.7
	614.736	167.07	8103		8.67e-001
KB1	9.572	0.12429	558	± 94	1.6
	680.018	175.07	8672		9.01e-001
KA1 Escape	6.897	0.00093	4	± 82	2.0
	497.353	151.63	6754		7.57e-001
KA2 Escape	6.874	0.00048	2	± 82	2.3
	495.782	151.42	6730		7.54e-001
KB1 Escape	7.830	0.00015	1	± 90	3.7
	561.064	160.20	8032		8.27e-001
15Br K			7258	± 660	
KA1	11.924	0.56311	4087	± 119	0.8
	840.626	193.35	8100		9.41e-001
KA2	11.878	0.29362	2131	± 99	1.0
	837.485	193.01	8134		9.47e-001
KB1	13.289	0.13261	963	± 85	1.4
	933.836	203.20	6901		9.57e-001
KB2	13.470	0.01003	73	± 82	1.9
	946.196	204.47	6710		9.58e-001
KA1 Escape	10.182	0.00037	3	± 92	2.5

	721.672	179.99	8446		9.17e-001
KA2 Escape	10.136	0.00020	1	± 92	2.0
	718.531	179.62	8464		9.16e-001
KB1 Escape	11.547	0.00006	0	± 92	2.7
	814.882	190.53	8389		9.43e-001

16Ag K			535	± 207	
KA1	22.163	0.65194	349	± 55	1.1
	1539.806		258.25		1865
KA2	21.990	0.34795	186	± 48	1.1
	1527.992		257.29		1952
KA1 Escape	20.421	0.00007	0	± 52	5.9
	1420.851		248.41		2684
KA2 Escape	20.248	0.00004	0	± 52	3.3
	1409.038		247.41		2748

8.79e-001

8.85e-001

9.26e-001

9.30e-001

17Ag L			21443	± 1272	
L1M3	3.234	0.11071	2374	± 89	27.2
	247.222	111.84	6046		6.51e-002
L1M2	3.203	0.06168	1323	± 82	29.6
	245.105	111.44	6136		6.01e-002
L1N3	3.747	0.08968	1923	± 69	25.7
	282.252	118.22	3580		1.73e-001
L1M5	3.437	0.00275	59	± 72	61.3
	261.084	114.41	5226		1.02e-001
L2M4	3.151	0.23201	4975	± 121	45.6
	241.554	110.77	6303		5.29e-002
L2N4	3.520	0.05488	1177	± 73	68.3
	266.751	115.44	4870		1.20e-001
L2M1	2.806	0.00206	44	± 82	354.0
	217.995	106.22	6753		1.59e-002
L2N1	3.428	0.00251	54	± 73	53.9
	260.469	114.29	5294		1.01e-001
L3M5	2.984	0.33280	7136	± 154	89.8
	230.150	108.59	6643		3.21e-002
L3N5	3.348	0.09173	1967	± 83	18.2
	255.006	113.29	5615		8.54e-002
L3M1	2.634	0.00260	56	± 81	111.0
	206.250	103.87	6574		6.77e-003
L3N1	3.256	0.00516	111	± 77	29.0
	248.724	112.12	5998		6.88e-002
L1M3 Escape	1.492	0.00128	27	± 27	2.4
	128.268	86.72	738		1.71e-012
L1M2 Escape	1.461	0.00073	16	± 24	2.7
	126.151	86.21	591		2.97e-013
L1N3 Escape	2.005	0.00085	18	± 50	1.1
	163.298	94.81	2529		1.30e-005
L2M4 Escape	1.409	0.00276	59	± 23	3.5
	122.600	85.34	546		1.10e-014
L2N4 Escape	1.778	0.00057	12	± 35	1.4
	147.797	91.32	1212		1.06e-007
L3M5 Escape	1.242	0.00421	90	± 25	8.0
	111.196	82.49	622		4.60e-021
L3N5 Escape	1.606	0.00102	22	± 29	2.7
	136.052	88.59	860		3.57e-010

Filter Background